

# RecoverChain Roadmap

Twenty improvements grouped into eight milestones across three tracks — from one-line fixes to a relational-model rework.

Snapshot 2026-08-31    Milestones 8    Tracks near-term · structural · production

Baseline master @ dac1317



**Start here:** the first two items of **M1** — Excel model training and per-row risk category. The smallest change that makes the ML and the pipeline actually respond to the data you upload, rather than treating every file as a generic failed payment scored by the baseline.



M1

**Data fidelity**

SIZE M

RECOMMENDED FIRST

Goal — an uploaded subscription / invoice / checkout file drives the matching logic, and the model trains whatever the file format.

**Excel model training**

MLTrainingEngine in `application/ml_training.py` hardcodes `pd.read_csv`. Branch on extension. Today every `.xlsx` dataset silently falls back to the deterministic baseline — no model ever trains.

**Per-row risk category**

`generate_cases` in `api/dataset_router.py` hardcodes `RiskCategory.FAILED_PAYMENT`. Read an event-type column. Unlocks the checkout / subscription / invoice branches that already exist in the risk, diagnosis, and recommendation engines but are never reached from data.

**raw\_payload type sanitisation**

Coerce `Timestamp` / `Decimal` / `numpy` scalars to primitives before storing. A real datetime cell currently returns a 500 from `generate-cases`.

**Receivables diagnosis vocabulary**

Map `net_terms_pending`, `dispute`, `awaiting_approval`, `bank_rejected` onto the existing `MISSED_COMMITMENT` / `UNRESOLVED_DISPUTE` / `MANDATE_FAILURE` causes. Invoice datasets are ~65% "cause unclear" today.

**Configurable currency**

Replace the hardcoded `INR` — read from a `currency` column or an env var.

**M2**

## Trust & explainability

SIZE M

Goal — a user can explain every result and take the output elsewhere.

### LLM narrator in the pipeline

Behind GEMINI\_API\_KEY, off by default so the app stays offline. Today only evaluation/ uses the LLM hook; the "plain words" lines on each step are deterministic templates.

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### Export

Download cases and outcomes as CSV / XLSX from the Insights report.

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### Insights depth

Per-dataset breakdown and a trend over time, not just an all-time roll-up.

**M3**

## Workflow robustness

SIZE S-M

Goal — the guided run survives a refresh, scales, and lets you explore.

### Persist run state

Keep the workflow context across navigation and refresh. Today a refresh restarts at Step 1.

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### Step 7 case picker

Browse the other generated cases, not only the highest-value one.

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### Background case generation

Make generate-cases a tracked job with progress. Auto-train and commits are already batched; execute + verify still run inline per row.

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### Registry housekeeping

Sweep orphaned model files (test leftovers). The current prune covers datasets only.

M4

## Test & quality coverage

SIZE M

Goal — the frontend stops being a blind spot.

### Frontend tests

The 10-step workflow, the column-mapping logic, computeInsights. There are none today.

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### Backend gaps

Correlation edge cases, prune, the .xlsx read paths.

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### CI runs both suites

Frontend build + tests alongside the 205 backend tests.

M5

**First-class domain model**

SIZE L

THE MILESTONE

Goal — a relational model instead of JSON blobs.

**Entities**

Customer, Transaction / Payment, Invoice, Subscription, CheckoutSession, PromiseToPay as modelled objects.

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**Entity resolution + shared reference\_id**

So related events correlate — a retry of the same invoice becomes one case, not two.

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**Migrate queryable sub-objects out of JSON columns**

The lifecycle stages that need filtering / aggregation move into real tables.

blocks → M6, M7

M6

**Multi-file & relationships**

SIZE L

Goal — correlate across files, not one file per run.

**Dataset linking / joins**

Payments ↔ customers ↔ invoices on a shared key.

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**Relationship detection in the Dataset Lab**

Suggest the join keys the way column mapping is suggested today.

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**Step 5 becomes functional**

"Data connection" stops being a stated limitation.

depends on → M5

**M7 Real execution** SIZE L

Goal — it recovers money instead of simulating it.

**Real payment adapter(s)**

Replace MockExecutionAdapter — e.g. Stripe, Razorpay — behind the same interface.

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**Real comms adapter**

Email / SMS for the reminder and payment-link actions.

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**Provider webhooks as settlement truth**

Replace the SANDBOX\_SIMULATION verification source with real reconciliation.

depends on → M5

**M8 Multi-tenancy & hardening** SIZE L

Goal — deployable as a service.

**Tenant isolation**

Data, models, and policy config separated per tenant.

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**Per-tenant policy configuration**

Retry caps, cooldowns, thresholds, consent rules set per customer.

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**Operational hardening**

Rate limiting, secrets management, production observability.

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**Compliance**

Legal review, penetration test, billing.

depends on → M5, M7